

Self-hosted deep research agent architecture

PLANNED

Research Farm Almanac · Slimy Knowledge Base

RUN ID 2026-06-06-sample-self-hosted-deep-research-agent	SLUG sample-self-hosted-deep-research-agent	DEPTH deep
PRIORITY normal	MODEL none	RUNNER slimy-research-plan-run@0.1.0
SOURCES 0	CITATIONS 0	CREATED 2026-06-05

Placeholder / Demo Output

This run has not completed real research yet. The content below is a skeleton generated by the seed-to-run planner. No web sources were fetched, no model synthesis was performed, and no findings should be treated as verified.

Research Seed

```
---
type: research_topic
status: queued
priority: normal
depth: deep
output: presentation_pdf
title: Self-hosted deep research agent architecture
slug: sample-self-hosted-deep-research-agent
created_by: Gurth
created_at: 2026-06-05
```

audience: technical_owner

visibility: owner

tags:

- ai-agents
- nuc
- slimy-kb
- habitat

seed_kind: one_shot

question: "What is the best architecture for a Slimy-owned deep research system that writes proof-packed reports back into slimy-kb and displays them in Habitat?"

scope_notes: "Phase 1 sample seed only. Do not run research from this file yet."

constraints:

- "Use slimy-kb as source of truth."
- "Keep Habitat owner-gated."
- "Do not build runner or queue controls during Phase 1."

related_projects:

- "slimy-kb"
- "gh-tracker"
- "slimy-harness"

assigned_critter: ""

campaign: "research-farm-bootstrap"

claim_token: ""

claimed_at: ""

started_at: ""

completed_at: ""

supersedes: ""

superseded_by: ""

Question

What is the best architecture for a Slimy-owned deep research system that

writes proof-packed reports back into slimy-kb and displays them in Habitat?

What matters

- KB-native topics and outputs
- Habitat read-only viewer first
- Presentation PDF output later
- Proof burrows for sources/citations/critic notes
- Farm/guild/pet-habitat theme
- No office metaphor
- No public exposure

Out of scope (for this forage)

- Building the actual runner. Phase 1 only plants the contract.
- Wiring Hermes or the mathomhaus/guild MCP.
- Adding any Habitat UI routes.
- Installing new system packages or starting any watchers.

Constraints

- All artifacts stay under ``research/``.
- Proof burrows are immutable once ``completed``.
- The almanac PDF is optional until Phase 3; the index contract permits

`pdf_path: null` .

Acceptance for the harvest

- A ``report.md`` that names the chosen architecture, the data flow, and the owner-gating boundary.

- A ``critic.md`` that red-teams the chosen architecture against the "no public exposure" and "no office metaphor" rules.

- A ``citations.json`` that links every architectural decision to a source fetched into `fetched/` .

- A ``RESULT.md`` that links the burrow directory into

`research/indexes/index.json` with `status: completed` .

Research Plan: sample-self-hosted-deep-research-agent

This is a placeholder plan skeleton. No research execution has started.

Objective

(To be filled when research begins.)

Approach

(To be filled when research begins.)

Steps

- Define search queries
- Fetch and review sources
- Synthesize findings
- Write report
- Write critic notes

Report

Research Report: sample-self-hosted-deep-research-agent

This is a placeholder report skeleton. No research has been conducted yet. Do NOT mark this run as complete.

Status

PLANNED - awaiting research execution.

Findings

(No findings yet.)

Recommendations

(No recommendations yet.)

▢ Slides

Presentation Slides: sample-self-hosted-deep-research-agent

This is a placeholder slides skeleton. No research has been conducted yet.

Slide 1: Title

sample-self-hosted-deep-research-agent

Slide 2: Key Findings

(No findings yet.)

Slide 3: Recommendations

(No recommendations yet.)

Slide 4: Sources

(No sources fetched yet.)

▢ Sources

No real sources yet (only placeholders).

▢ Citations

No citations recorded yet.

▢ Critic Notes

Critic Notes: sample-self-hosted-deep-research-agent

This is a placeholder critic skeleton. No research has been conducted yet.

Strengths

(To be evaluated after research.)

Weaknesses

(To be evaluated after research.)

Red-Team Assessment

(To be evaluated after research.)

□ Proof Burrow / RESULT

```
# Result
```

```
RESULT=PLANNED
```

```
RUN_ID=2026-06-06-sample-self-hosted-deep-research-agent
```

```
## What this means
```

This is a Phase 2 lifecycle skeleton created by the seed-to-run planner.

- No web research has been run yet.
- No sources fetched yet.
- No report has been written yet.
- No PDF has been generated yet.
- No model was used for research.

This run folder exists only as a skeleton ready for a future research execution phase.

```
## Lifecycle
```

- Status: planned
- Created by: slimy-research-plan-run@0.1.0
- This RESULT.md will be updated when research is actually executed.

Tags

- ai-agents
- nuc
- slimy-kb
- habitat